2 ALARM NUMBERS AND BRIEF DESCRIPTIONS

2.1 SERVO ALARM

Alarm No.	LED SV	Description
SV0001		SYNC ALIGNMENT ERROR
SV0002		SYNC EXCESS ERROR ALARM 2
SV0003		SYNCHRONOUS/COMPOSITE/SUPERIMPOSED CONTROL MODE CAN'T BE CONTINUED
SV0004		EXCESS ERROR (G31)
SV0005		SYNC EXCESS ERROR (MCN)
SV0006		ILLEGAL TANDEM AXIS
SV0007		SV ALM ANOTHER PATH(MULTI AMP.)
SV0010		SV OVERHEAT
SV0011		SV MOTOR OVER CURRENT(SOFT)
SV0012	4	SV DRIVE OFF CIRCUIT FAILURE
SV0013	3	SV CPU BUS FAILURE
SV0014	J	SV CPU WATCH DOG
SV0015	2 (Blinking)	SV LOW VOLT DRIVER
SV0016	b. C. d.	SV CURRENT DETECT ERROR
SV0017	11	SV INTERNAL SERIAL BUS FAILURE
SV0018	11 (Blinking)	SV ROM DATA FAILURE
SV0019	b C d (Blinking)	SV GROUND FAULT
SV0024		PS SOFT THERMAL
SV0031		PS ILLEGAL PARAMETER
SV0032		PS CONTROL AXIS ERROR 1
SV0033		PS CONTROL AXIS ERROR 2
SV0034		PS HARDWARE ERROR
SV0035	—	SV NO FAILURE
SV0036	A (Blinking)	PHASE OPEN
SV0037	9 (Blinking)	FAILURE OF SV(OPEN)
SV0038	—	FAILURE OF CURRENT CTRL.
SV0039	8 (Blinking)	FAILURE OF SV(SHORT)
SV0040		PS EXT-COMP. ERROR
SV0041		PS PFB-R ERROR
SV0042		PS PFB-C ERROR
SV0043		PS SUB MODULE ERROR
SV0044		MISMATCHED FUNCTION CODE

Alarm No.	LED	Description
	SV	
SV0301		APC ALARM: COMMUNICATION ERROR
SV0302		APC ALARM: OVER TIME ERROR
SV0303		APC ALARM: FRAMING ERROR
SV0304		APC ALARM: PARITY ERROR
SV0305		APC ALARM: PULSE ERROR
SV0306		APC ALARM: OVER FLOW ERROR
SV0307		APC ALARM: MOVEMENT EXCESS ERROR
SV0360		ABNORMAL CHECKSUM(INT)
SV0361		ABNORMAL PHASE DATA(INT)
SV0363		ABNORMAL CLOCK(INT)
SV0364		SOFT PHASE ALARM(INT)
SV0365		BROKEN LED(INT)
SV0366		PULSE MISS(INT)
SV0367		COUNT MISS(INT)
SV0368		SERIAL DATA ERROR(INT)
SV0369		DATA TRANS. ERROR(INT)
SV0380		BROKEN LED(EXT)
SV0381		ABNORMAL PHASE (EXT)
SV0382		COUNT MISS(EXT)
SV0383		PULSE MISS(EXT)
SV0384		SOFT PHASE ALARM(EXT)
SV0385		SERIAL DATA ERROR(EXT)
SV0386		DATA TRANS. ERROR(EXT)
SV0387		ABNORMAL ENCODER(EXT)
SV0401		IMPROPER V_READY OFF
SV0403		CARD/SOFT MISMATCH
SV0404		IMPROPER V_READY ON
SV0407		EXCESS ERROR
SV0409		DETECT ABNORMAL TORQUE
SV0410		EXCESS ERROR (STOP)
SV0411		EXCESS ERROR (MOVING)
SV0413		LSI OVERFLOW
SV0415		MOTION VALUE OVERFLOW
SV0417		ILL DGTL SERVO PARAMETER
SV0420		SYNC TORQUE EXCESS
SV0421		EXCESS ERROR(SEMI-FULL)
SV0422		EXCESS VELOCITY IN TORQUE
SV0423		EXCESS ERROR IN TORQUE
SV0430		SV MOTOR OVERHEAT
SV0431		PS OVERLOAD
SV0432		PS LOW VOLT. CONTROL
SV0433		PS LOW VOLT. DC LINK
SV0434	2	SV LOW VOLT CONTROL
SV0435	5	SV LOW VOLT DC LINK
SV0436		SOFTTHERMAL(OVC)
SV0437		PS OVERCURRENT
SV0438	b C d	SV ABNORMAL CURRENT
SV0439		PS OVER VOLT. DC LINK

Alarm No.	LED SV	Description
SV0441		ABNORMAL CURRENT OFFSET
SV0442		PS PRE-CHARGE FAILURE
SV0443		PS INTERNAL FAN FAILURE
SV0444	1	SV INTERNAL FAN FAILURE
SV0445		SOFT DISCONNECT ALARM
SV0446		HARD DISCONNECT ALARM
SV0447		HARD DISCONNECT(EXT)
SV0448		UNMATCHED FEEDBACK ALARM
SV0449	8. 9. A.	SV IPM ALARM
SV0453		SPC SOFT DISCONNECT ALARM
SV0454		ILLEGAL ROTOR POS DETECT
SV0456		ILLEGAL CURRENT LOOP
SV0458		CURRENT LOOP ERROR
SV0459		HI HRV SETTING ERROR
SV0460		FSSB DISCONNECT
SV0462		SEND CNC DATA FAILED
SV0463		SEND SLAVE DATA FAILED
SV0465		READ ID DATA FAILED
SV0466		MOTOR/AMP. COMBINATION
SV0468		HI HRV SETTING ERROR(AMP)
SV0474		EXCESS ERROR (STOP:SV)
SV0475		EXCESS ERROR(MOVE:SV)
SV0476		ILLEGAL SPEED CMD.(SV)
SV0477		ILLEGAL MACHINE POS.(SV)
SV0478		ILLEGAL AXIS DATA(SV)
SV0481		SAFETY PARAM ERROR(SV)
SV0484		SAFETY FUNCTION ERROR(SV)
SV0488		SELF TEST OVER TIME
SV0489		SAFETY PARAM ERROR(CNC)
SV0490		SAFETY FUNCTION ERROR(CNC)
SV0494		ILLEGAL SPEED CMD.(CNC)
SV0496		ILLEGAL AXIS DATA(CNC)
SV0498		AXIS NUMBER NOT SET(CNC)
SV0600	8	SV DC LINK OVER CURRENT
SV0601	F	SV EXTERNAL FAN FAILURE
SV0602	6	SV RADIATOR OVERHEAT
SV0603	8. 9. A.	INV. IPM ALARM(OH)
SV0604	P	AMP COMMUNICATION ERROR
SV0606		PS EXTERNAL FAN FAILURE
SV0607		PS IMPROPER INPUT POWER
SV0646		ABNORMAL ANALOG SIGNAL(EXT)
SV0652		TEMP.ERROR
SV0654	7	DB RELAY FAILURE
SV1025		V_READY ON (INITIALIZING)
SV1026		ILLEGAL AXIS ARRANGE
SV1055		ILLEGAL TANDEM AXIS
SV1067		FSSB:CONFIGURATION ERROR(SOFT)

2.2 SPINDLE ALARM

Alarm No.	LED SP	Description
SP9001	01	MOTOR OVERHEAT
SP9002	02	EX DEVIATION SPEED
SP9003	03	DC-LINK FUSE IS BROKEN
SP9004	04	PS IMPROPER INPUT POWER
SP9006	06	THERMAL SENSOR DISCONNECT
SP9007	07	OVER SPEED
SP9009	09	OVERHEAT MAIN CIRCUIT
SP9010	10	LOW VOLT INPUT POWER
SP9011	11	PS OVER VOLT. DC LINK
SP9012	12	OVERCURRENT POWER CIRCUIT
	13	CPU DATA MEMORY FAULT
SP9014	14	SOFTWARE SERIES MISMATCH
SP9015	15	SPINDLE SWITCHING FAULT
SP9016	16	RAM ERROR
SP9017	17	ID NUMBER PARITY ERROR
	18	SUMCHECK ERROR PROGRAM ROM
SP9019	19	EXCESS OFFSET CURRENT U
SP9020	20	EXCESS OFFSET CURRENT V
SP9021	21	POS SENSOR POLARITY ERROR
SP9022	22	SP AMP OVER CURRENT
	24	SERIAL TRANSFER ERROR
SP9027	27	DISCONNECT POSITION CODER
SP9029	29	OVERLOAD
SP9030	30	PS OVERCURRENT
SP9031	31	MOTOR LOCK
SP9032	32	SIC-LSI RAM FAULT
SP9033	33	PS PRE-CHARGE FAILURE
SP9034	34	ILLEGAL PARAMETER
SP9036	36	OVERFLOW ERROR COUNTER
SP9037	37	ILLEGAL SETTING VELOCITY DETECTOR
SP9041	41	ILLEGAL 1REV SIGN OF POSITION CODER
SP9042	42	NO 1REV SIGN OF POSITION CODER
SP9043	43	DISCONNECT POSITION CODER DEF. SPEED
SP9046	46	ILLEGAL 1REV SIGN OF SCREW CUT
SP9047	47	ILLEGAL SIGNAL OF POSITION CODER
SP9049	49	DEF. SPEED IS OVER VALUE
SP9050	50	SYNCRONOUS VALUE IS OVER SPEED
SP9051	51	PS LOW VOLT. DC LINK
SP9052	52	ITP FAULT 1
SP9053	53	ITP FAULT 2
SP9054	54	OVERCURRENT
SP9055	55	ILLEGAL POWER LINE
SP9056	56	COOLING FAN FAILURE
SP9058	58	PS OVERLOAD
SP9059	59	PS INTERNAL FAN FAILURE
SP9061	61	SEMI-FULL ERROR EXCESS
SP9065	65	SERIAL SPINDLE ALARM
SP9066	66	COM. ERROR BETWEEN SP AMPS
SP9067	67	FSC/EGB COMMAND ERROR
SP9068	68	ILLEGAL SPINDLE PARAMETER
SP9069	69	SAFETY SPEED OVER

Alarm No.	LED	Description
	SV	
SV1068		DUAL CHECK SAFETY ALARM
SV1069		EXCESS ERROR(SERVO OFF:CNC)
SV1070		EXCESS ERROR(SERVO OFF:SV)
SV1071		EXCESS ERROR(MOVE:CNC)
SV1072		EXCESS ERROR(STOP:CNC)
SV1100		S-COMP. VALUE OVERFLOW
SV5134		FSSB:OPEN READY TIME OUT
SV5136		FSSB:NUMBER OF AMP. IS INSUFFICIENT
SV5137		FSSB:CONFIGURATION ERROR
SV5139		FSSB:ERROR
SV5197		FSSB:OPEN TIME OUT
SV5311		FSSB:ILLEGAL CONNECTION

Alarm No.	LED SP	Description
SP9070	70	ILLEGAL AXIS DATA
SP9071	71	SAFETY PARAMETER ERROR
SP9072	72	MISMATCH RESULT OF MOTOR SPEED CHECK
SP9073	73	MOTOR SENSOR DISCONNECTED
SP9074	74	CPU TEST ERROR
SP9075	75	CRC ERROR
SP9076	76	INEXECUTION OF SAFETY FUNCTIONS
SP9077	77	MISMATCH RESULT OF AXIS NUMBER CHECK
SP9078	78	MISMATCH RESULT OF SAFETY PARAMETER CHECK
SP9079	79	INITIAL TEST ERROR
SP9080	80	ALARM AT THE OTHER SP AMP.
SP9081	81	1-ROT MOTOR SENSOR ERROR
SP9082	82	NO 1-ROT MOTOR SENSOR
SP9083	83	MOTOR SENSOR SIGNAL ERROR
SP9084	84	SPNDL SENSOR DISCONNECTED
SP9085	85	1-ROT SPNDL SENSOR ERROR
SP9086	86	NO 1-ROT SPNDL SENSOR
SP9087	87	SPNDL SENSOR SIGNAL ERROR
SP9088	88	COOLING RADI FAN FAILURE
SP9089	89	SUB MODULE SM (SSM) ERROR
SP9090	90	UNEXPECTED ROTATION
SP9091	91	POLE POSITION COUNT MISS
SP9092	92	OVER SPEED TO VELOCITY COMMAND
	A	Program ROM error
	A1	Program ROM error
	A2	Program ROM error
SP9110	b0	AMP COMMUNICATION ERROR
SP9111	b1	PS LOW VOLT. CONTROL
SP9113	b3	PS EXTERNAL FAN FAILURE
SP9114	b4	PS CONTROL AXIS ERROR 1
SP9115	b5	PS CONTROL AXIS ERROR 2
SP9120	C0	COMMUNICATION DATA ERROR
SP9121	C1	COMMUNICATION DATA ERROR
SP9122	C2	COMMUNICATION DATA ERROR
SP9123	C3	SPINDLE SWITCH CIRCUIT ERROR
SP9124	C4	LEARNING CTRL ROTATION CMD ERROR
SP9125	C5	LEARNING CTRL COMP ORDER ERROR
SP9127	C7	LEARNING CTRL PERIOD ERROR
SP9128	C8	SP SYNC VELOCITY ERROR EXCESS
SP9129	C9	SP SYNC POSITION ERROR EXCESS
SP9130	d0	TORQUE TANDEM POLARITY ERROR
SP9131	d1	SPINDLE TUNING FUNCTION ALARM
SP9132	d2	SER.SENSOR DATA ERROR
SP9133	d3	SER.SENSOR TRANSFER ERROR
SP9134	d4	SER.SENSOR SOFT PHASE ERROR
SP9137	d7	SP DEVICE COMMUNICATION ERROR
SP9138	d8	CURRENT LIMIT SETTING ERROR
SP9139	d9	SER.SENSOR PULSE MISS
SP9140	E0	SER.SENSOR COUNT MISS
SP9141	E1	SER.SENSOR NO 1-ROT SIGNAL
SP9142	E2	SER.SENSOR ABNORMAL
SP9143	E3	CS HIGH SPEED CHANGE CMD ERROR
SP9144	E4	CURRENT DETECT CIRCUIT ERROR

Alarm No.	LED	Description
	SP	
SP9145	E5	LOW VOLTAGE DRIVER
SP9148	E8	AXIS NUMBER NOT SET
SP9153	F3	SP NO FAILURE
SP9154	F4	PHASE OPEN
SP9155	F5	FAILURE OF SP (OPEN)
SP9156	F6	FAILURE OF CURRENT CTRL.
SP9157	F7	FAILURE OF SP (SHORT)
SP9158	F8	MODE MISMATCH (DCS)
SP9159	F9	MISMATCHED FUNCTION CODE
SP9160	G0	THERMISTOR DISCONNECTION
SP9161	G1	POW. CABLE SHORT CIRCUIT
SP9162	G2	DELAY OF SP POS. DATA
SP9163	G3	ILLEGAL SP POS. DATA
SP9164	G4	FAILURE OF MASTER SENSOR
SP9165	G5	CONTROL METHOD CHANGED
	UU	FSSB master port wire break
	LL	FSSB slave port wire break
SP0756		ILLEGAL AXIS DATA
SP9204	M4	PS SOFT THERMAL
SP9211	L1	PS ILLEGAL PARAMETER
SP9212	L2	PS HARDWARE ERROR
SP9213	L3	PS EXT-COMP. ERROR
SP9214	L4	PS PFB-R ERROR
SP9215	L5	PS PFB-C ERROR
SP9216	L6	PS SUB MODULE ERROR